

18.4 SpikeRAM: A 48.1pW/Synapse/Bit Event-Driven Spiking Compute-Near/In-Memory Processor with Neuromorphic Sensor Enabling Life-Long On-Chip Learning

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Abstract

SpikeRAM is a high efficiency (48.1pW/Synapse/Bit) memory-centric neuromorphic system with perception, computing and on-chip learning, achieving 464M synapses and 8.28mW power in real-time processing, while realizing few-shot learning for event-based signature

verification. The EVS-sCNN core enables asynchronous sensing and feature extraction. The sFC-OCL core enables event-driven inference and learning with the e-OTBP algorithm. Gray-code weights and ternary gradients reduce memory write times by over 86%.

Edge perceptual SoCs increasingly demand high energy efficiency, privacy and adaptability for real-world tasks as shown in Fig. 18.4.1 (top) [1–6]. For perception, an event-based vision sensor (EVS) with asynchronous, sparse and edge-only imaging, offers low power and high privacy [7–8]. For computation, the use of a spiking neural network (SNN) further increases energy efficiency through the spike-driven paradigm [9–14]. In addition, on-chip learning (OCL) with emerging non-volatile memories (eNVMs) enables permanent in-situ training without cloud computing, mitigating privacy risks while supporting user-specific and environment adaptation [15–17]. However, to achieve energy efficiency, privacy and adaptability simultaneously, there are three challenges as shown in Fig. 18.4.1 (middle). 1) Inefficient inference arises from data movement due to the separation of sensing, memory and computing, along with frame conversion and redundant multiply-accumulations. 2) High power and memory overhead in spatial-temporal OCL, which scales with the depth of time-window (TW). 3) Excessive memory updates during OCL increase energy and hamper lifetimes for eNVMs with limited writing endurance [18–21].

To tackle these challenges, we introduce the SpikeRAM system comprising an on-chip EVS-integrated spiking convolution neural network (EVS-sCNN) core with near-memory computing (NMC) and a spiking fully connected OCL (sFC-OCL) core featuring compute-in-memory (CIM). Fig. 18.4.1 (bottom) depicts three key features of SpikeRAM: 1) Memory-centric asynchronous near-sensor computing. The SNN core directly processes asynchronous events from on-chip EVS, eliminating frame conversion, reducing data movement and redundant operations. 2) Eligibility one-time backpropagation (e-OTBP) OCL algorithm trains a multi-layer network through a single TW and eligibility trace (ET). It maintains low power, while significantly reducing memory requirements by 73.1% at 32 TWs. 3) Ternary gradients with gray-code weights significantly reduce memory programming times. Combined with the high convergence of e-OTBP, it enhances both durability and energy efficiency. On MNIST and DVS Gesture, the average programming times are reduced by 89.1% and 88.4%, respectively. To summarize, SpikeRAM is an EVS-based spiking sense-memory-compute system with OCL. It also enables eNVM-based in-situ training and can extend eNVM lifetime by over 10x.

Figure 18.4.2 (top) illustrates the architecture of the SpikeRAM system. In the EVS-sCNN core, raw events from on-chip EVS (3.8MEps and 130.9dB dynamic range) or external sources are denoised and routed to the sCNN processing elements (PEs) for feature extraction through configurable convolution layers, scheduled by a Network-on-Chip (NoC) for diverse tasks. In the sFC-OCL core, extracted feature events are processed through sFC and OCL PEs with configurable filters and neurons for inference and OCL. Fig. 18.4.2 (bottom) shows the workflow of the sCNN PE, which adopts a fully asynchronous design with dual-rail handshake logic to eliminate the clock tree overhead. Each PE performs event-driven convolution, activating kernels only upon event arrival and skipping blank regions to avoid invalid operations, enabling sparsity adaptation. During processing, padding ensures dimensional consistency, and the kernel anchor determines output feature locations. Address sweep generates addresses for Leaky-Integrate-and-Fire (LIF) neurons, where non-zero 8b weights are accumulated into 16b membrane potential (V_{mem}). Once V_{mem} exceeds the threshold, neurons reset and output events. These events are merged via polling, packetized, and routed to the next layer or the sFC-OCL core under NoC control.

Figure 18.4.3 (top) illustrates the system diagram of the CIM-based sFC-OCL core. The first sFC PE contains 8 digital CIM macros, each including a 32Kb RAM, containing 8b weights with sense amplifiers (SAs), gray-code decoders, parallel shift-adders, and 16 configurable neurons. The second sFC PE consists of 10 CIM macros with a total of 10Kb for weights and 10 output neurons. Within each TW, input events are compressed into weighted spikes and routed to a configurable event filter to exploit input sparsity, skipping up to 92.4% of redundant computations in DVS Gesture. Addresses of weighted spikes activate the corresponding word-lines in the first sFC PE and decoded weights are shifted and accumulated into V_{mem} to avoid multiplications. Neurons support LIF, IF and InoF behaviors, allowing dynamic trade-offs between temporal sensitivity and energy efficiency. The generated spikes are routed to the second sFC-PE for subsequent processing. At the end of inference, the OCL PEs based on e-OTBP are activated for backpropagation by reusing the preserved inference states. Fig. 18.4.3 (bottom) shows a handshake-based pipeline that ensures efficient inference and learning in the sFC-OCL core.

Figure 18.4.4 (top) shows the e-OTBP algorithm and ternary gradient in OCL. The e-OTBP performs global learning through a single TW and captures temporal information via both ETs and output V_{mem} across previous TWs, retained in inference and reused in learning. Gradients of layer 2 are computed by errors and corresponding input spikes. Gradients of layer 1 are computed by propagating errors via transposed weights and a surrogate (Surr) function, then combined with ETs. Both gradients are fed into ternary quantization function yielding the ternary value. The computing graph shows that e-OTBP eliminates historical computations compared with conventional backpropagation through time (BPTT) algorithm. For efficient hardware implementation, the Surr function is binarized using two comparators, the ET cell is realized via a cyclic accumulator, and the FC weight transpose is computed via a horizontal adder tree to reuse the weight array. Compared with BPTT, the memory and computation overheads of e-OTBP are 4.3% and 3.1%, respectively.

Figure 18.4.4 (bottom) illustrates the gray-code dataflow. All weights are stored in the array using gray-code encoding. During inference, gray-code decoders implemented by parallel XOR logic are integrated with the SAs to enable binary readout of weights for neuron accumulations. During learning, readout weights are added or subtracted with ternary gradients and updated in gray-code. This combination ensures that each weight is updated by no more than 1b per batch, significantly reducing memory programming times. SpikeRAM can be either standalone or combined with a non-volatile MRAM array to achieve large-scale neural networks and permanent learning. With an external 4Mb MRAM chip, the parameters can be expanded by 5.4x for 101-class N-Caltech101 [23] tasks through time multiplexing.

Figure 18.4.5 (top) shows the event-based signature verification system as a demonstration of real-world private edge learning. Signatures are private data for authentication, leading to cloud training privacy leakage issues. Conventional image-based methods ignore temporal dynamics, reducing accuracy and raising forgery risks. SpikeRAM addresses these by sensing the spatial-temporal features with EVS, followed by SNN-based inference and OCL. Fig. 18.4.5 (top left) shows the system setup. The on-chip EVS with an infrared filter captures the signature signals from an infrared pen at 200Hz within the 850nm band. To suppress interference from sensor and ambient noise, the NeuroPass layer employs a LIF-based sCNN to selectively respond to a specific frequency range, thereby improving the signal-to-noise ratio as shown in Fig. 18.4.5 (top middle). SpikeRAM extracts spatial features while preserving sign speed thanks to the high temporal resolution of EVS as shown in Fig. 18.4.5 (right). Although forgeries appear nearly same in static image, they are easily distinguishable in temporal domain. Based on SpikeRAM, we introduce the EHSV dataset, comprising one class of genuine signatures and four classes of forgeries. With 2-shot OCL from random initialization, SpikeRAM achieves 94.3% accuracy in authenticity verification and classification. Fig. 18.4.5 (middle) shows the measurement results. Static power and dynamic power for real-time processing are 1.26mW and 8.28mW, respectively. On MNIST, DVS Gesture and DailyDVS, SpikeRAM achieves rapid convergence from random initialization, reaching 96.4%, 92.9% and 90.5% accuracy, respectively, with only 1 OCL epoch.

Figure 18.4.5 (middle right and bottom) shows that integrating ternary gradients, gray-code weights, and the high convergence of e-OTBP significantly reduces MRAM write operations. In DVS Gesture, mean and standard deviation of the update times are reduced by 10.9x and 35.6x, respectively. Across four datasets, the maximum number of memory devices written is reduced by 92.8% and the total write time per sample is reduced by 86.3%, on average. These results indicate that SpikeRAM effectively mitigates MRAM endurance limitations during OCL.

Figure 18.4.6 compares state-of-the-art neuromorphic processors. SpikeRAM improves power density (48.1pW/Synapse/Bit) by 2.7x over [2] through its NMC and CIM hybrid architecture with asynchronous and event-driven sparsity adaptation. SpikeRAM supports 464M synapses and 328K neurons for diverse networks, improving network density (441.9Synapse/Byte) by 24.3x over [10]. Learning efficiency is evaluated by the total energy per learning task. SpikeRAM achieves outstanding learning efficiency and accuracy across multiple event-based tasks, validated with on-chip SRAM and an external 40nm, 4Mb MRAM chip. The MRAM lifetime is extended by 13.9x in OCL, while energy consumption remains

comparable. At the system level, SpikeRAM integrates on-chip EVS for real-time sensing, processing and learning, achieving 8.28mW system power in real-world tasks. Fig. 18.4.7 shows the summaries, voltage characteristics, micrographs and test system. SpikeRAM enables both real-time perception and inference, while supporting long-life learning in an eNVM-based system.

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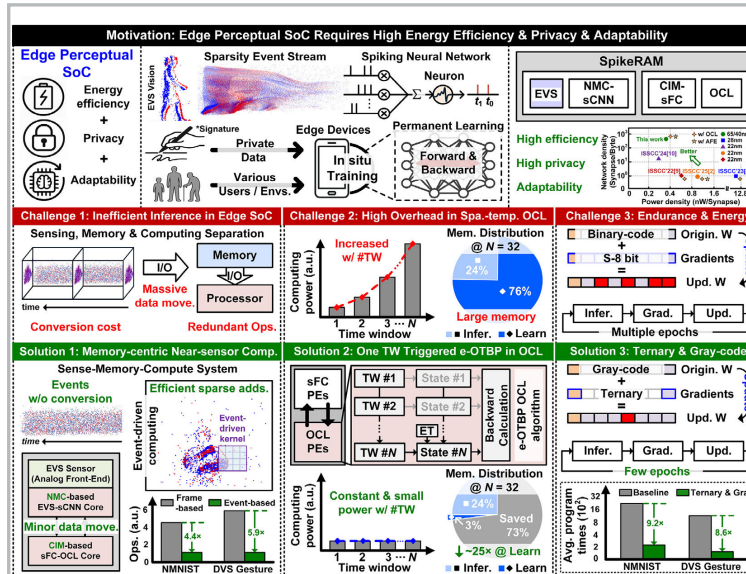


Figure 18.4.1: Motivation of edge perceptual SoC with high energy efficiency, privacy and adaptability (top), challenges (middle), and our solutions in SpikeRAM (bottom).

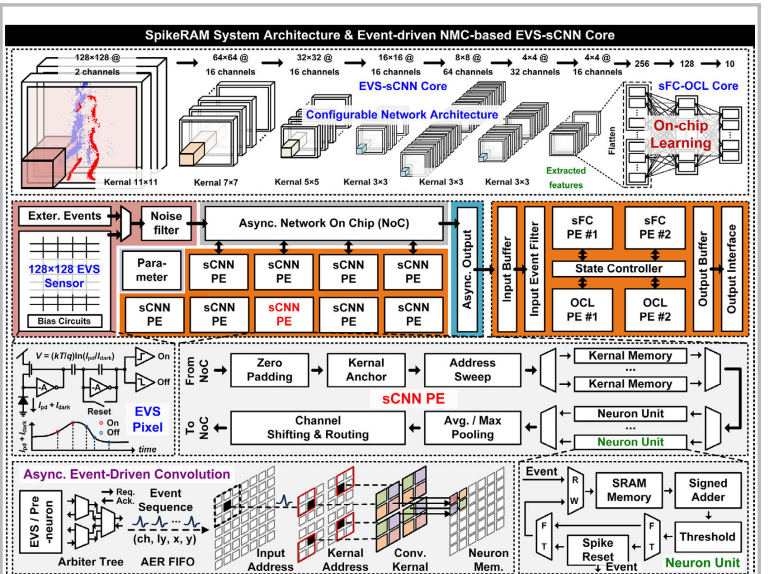


Figure 18.4.2: Architecture of SpikeRAM system (top), and event-driven convolution workflow in EVS-sCNN core (bottom).

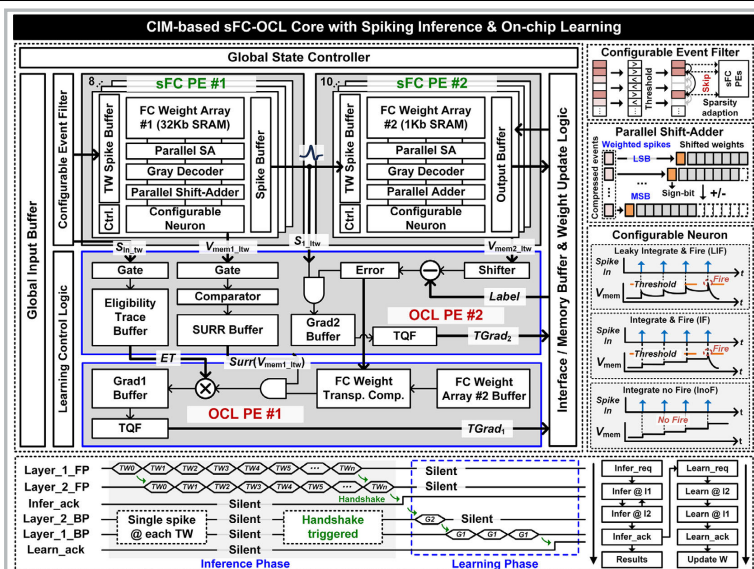


Figure 18.4.3: System diagram of sFC-OCL core (top), event filter, shift-adder with weighed spikes and configurable neurons (top right), and handshake-based pipeline (bottom).

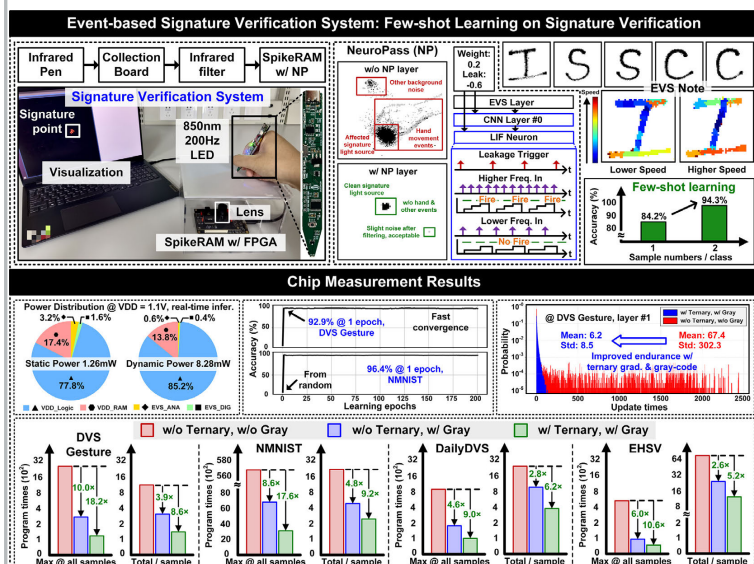


Figure 18.4.5: Event-based signature verification system based on SpikeRAM for few-shot on-chip learning (top), and chip measurement results (bottom).

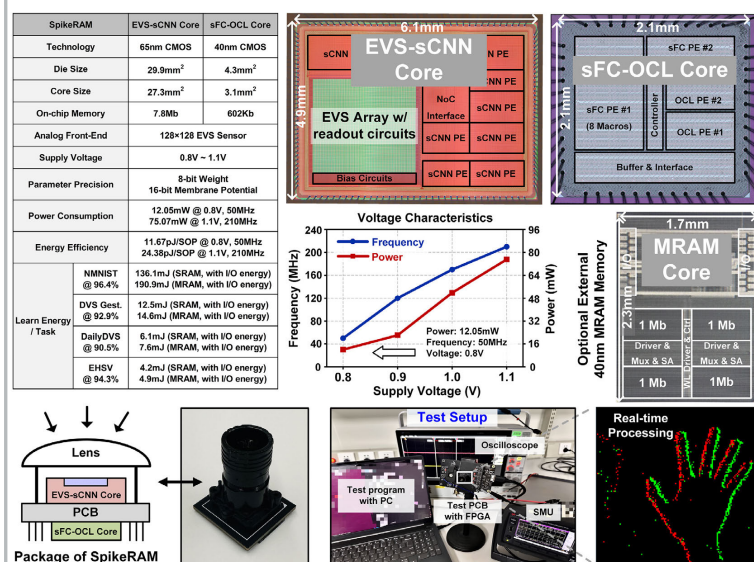


Figure 18.4.7: Summary table (top left), voltage characteristics and chip micrograph (top right), and measurement system (bottom).

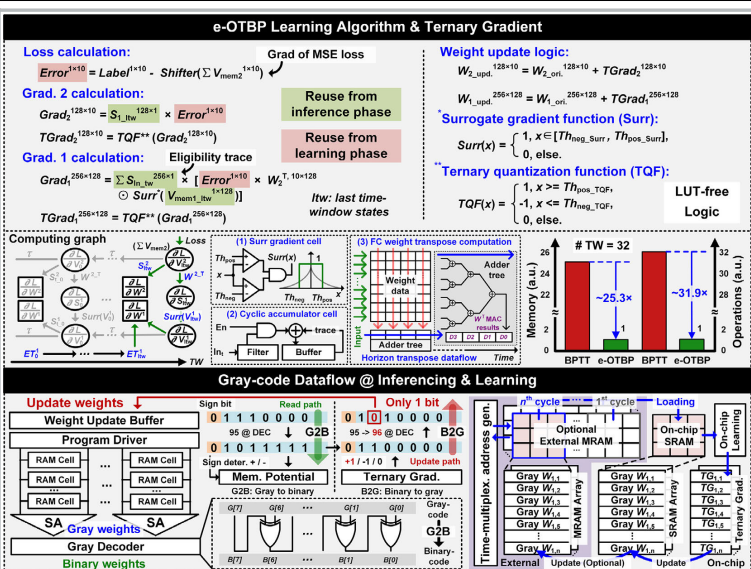


Figure 18.4.4: e-OTBP on-chip learning algorithm and ternary gradient implementations (top), and gray-code dataflow with weight updating and optional learning in MRAM (bottom).

	ISSCC'22 [9]	ISSCC'23 [4]	ISSCC'24 [10]	JSSC'25 [5]	ISSCC'25 [2]	This work
Technology	28nm	28nm	28nm	28nm	22nm	65nm / 40nm
On-chip Learning	Mod. stoch. e-prop	✓	✓	✓	✓	e-OTBP
Computation Mode	Digital	Digital	CIM	Digital	Digital	NMC / CIM
Architecture	sRNN	sFC	sCNN	CNN / sCNN	sFC	sCNN
Analog Front-End	X	X	X	X	16-ch Gas Sensor	128x128 EVS Sensor
Task	Hand gest. classif. Keyword spotting	Image classif. Hand gest. classif.	Image classif. Hand gest. classif.	Image classif. 3-D sample classif.	Gas classif.	Image classif. Hand gest. classif. Human action classif. Event handling signature verif.
Datasets & Input feature size	DVS Gest.: 16x16 SHD: 234x1 DCI: 40x1	MNIST: 17x17 DVS Gest.: 16x16 N-TIDIGIT: 64x8 SeNci: 8x128	N.A.	EMNIST: 28x28 ModelNet40: N.A.	N.A.	MNIST ¹ : 34x34 (raw data) DVS Gest. ² : 128x128 (raw data) DailyDVS ³ : 128x128 (raw data) EHSV ⁴ : 128x128 (raw data)
Accuracy with On-chip Learning	DVS Gest.: 87.3% SHD: 90.7% DCI: 96.4%	MNIST: 96.0% DVS Gest.: 92.0% N-TIDIGIT: 92.6% SeNci: 95.7%	/	EMNIST: 72.67% ModelNet40: 89.60%	FTs: 98.23% UCI VOCs: 98.0% LCA: 97.7%	MNIST: 96.4% DVS Gest.: 92.9% DailyDVS: 89.9% EHSV: 94.3%
Memory Size	138Kb	266.5Kb	110Kb	121.665Mb	58Kb	1.00Mb
Supported # of Synapse	132K	258K	2M	4.450	53K	464M ⁵
Energy Efficiency	5.30pJ/SOP @ 0.5V, 13MHz	1.49pJ/SOP @ 0.56V, 40MHz	3.78pJ/SOP @ 0.55V, 51MHz	5.187pJ/SOP @ 0.625V, 1200MHz	N.A.	11.67pJ/SOP ⁶ (System level) @ 0.8V, 50MHz, 80W
System Power	79uW	2.91mW	524uW	0.19W	55.12uW	12.05mW @ 0.8V, 50MHz, 6 sCNN & 2 sFC layers 8.25mW @ 1.1V, 10MHz, real-time infer. w/ scope
FoM: Power Density ⁷	0.6mW/Synapse	11.2mW/Synapse	0.26mW/Synapse	N.A.	0.13mW/Synapse/Bit	0.36mW/Synapse (8-bit) 0.04mW/Synapse/Bit
FoM: Network Density ⁸	0.96 Synapses/Byte	0.97 Synapses/Byte	18.18 Synapses/Byte	N.A.	0.91 Synapses/Byte	441.90 Synapses/Byte
Learn Energy / Task ⁹	DVS Gest.: 10.5J SHD: 7.4J DCI: > 3.4uJ	MNIST: 489.6mJ DVS Gest.: 11.2mJ N-TIDIGIT: 14.3mJ SeNci: 12.2uJ	/	N.A.	N.A.	MNIST: 136.1mJ DVS Gest.: 12.5mJ DailyDVS: 6.1mJ EHSV: 4.2mJ

Figure 18.4.6: Comparison with state-of-the-art neuromorphic processors.



Figure 18.4.7: Summary table (top left), voltage characteristics and chip micrograph (top right), and measurement system (bottom).